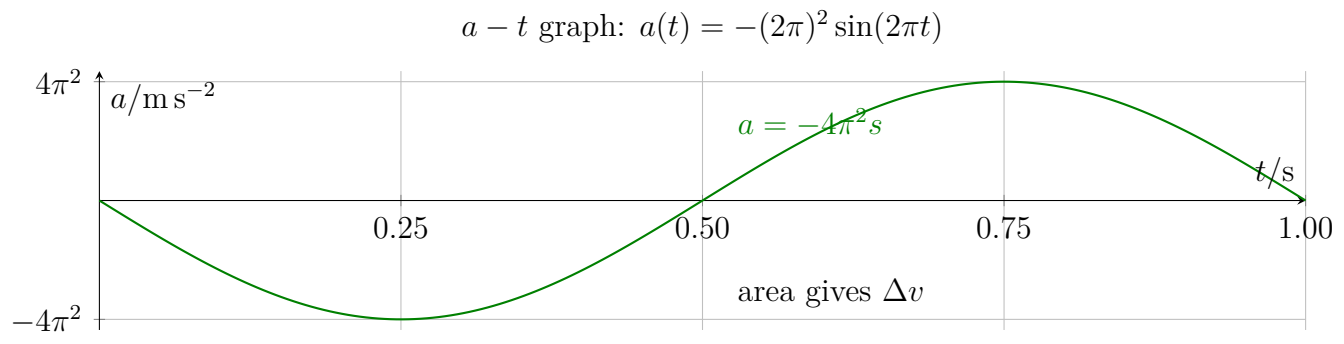
$a - t$ graph: $a(t) = -(2\pi)^2 \sin(2\pi t)$
$4\pi^2$
$a/\mathrm{m\,s^{-2}}$
$a = -4\pi^2 s$
$t/\mathrm{s}$
0.25
0.50
0.75
1.00
area gives $\Delta v$
$-4\pi^2$